

Portfolio Question 4.1.P2 - How many Eigenvalues to retain?

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Portfolio 4.1.P2 (v4.0.0)

Portfolio Question 4.1.P2 - How many Eigenvalues to retain?

- ▶ How many dimensions should we retain in an Eigenvalue decomposition?
 - ▶ Is there an objective solution? Under what circumstances?

How many PCs?

- ▶ We select only those PCs whose eigenvalues are “large enough”. Procedures include:
 - ▶ **scree plot**, looking for an elbow in the distribution
 - ▶ **EVs > 1** , justified by random graph theory
 - ▶ **Tracey-Widom theory**, similarly
 - ▶ **Horn’s criterion**, based on simulating random matrices for the remaining matrix structure after K are chosen
 - ▶ **Velicer’s MAP criterion**, similar
- ▶ In practice any can be “wrong”, so common sense should be applied.
- ▶ Eigenvectors associated with Eigenvalues that are “too small” will contain some noise, even if they still contain a signal.

How many PCs?

- ▶ In the data from the lecture, we have only 4 features. And we mean centred, which removes a degree of freedom. So the data should lie on a 3D subspace:

```
> round(testdata.eigen$values[1:4],8)
[1] 1859.25164 124.70157 16.04679 0.00000
> round(((testdata.svd$d)^2)/3,8)
[1] 1859.25164 124.70157 16.04679 0.00000
```

- ▶ Scaling by the number of features is not important for the SVD when computing the *proportion* of variance explained. Just square the singular values.

Mathematical properties:

- ▶ If X is **positive definite**, its eigenvalues are > 0 .
- ▶ If X is **positive semi-definite**, its eigenvalues are ≥ 0 .
- ▶ positive definite: If $\forall v \neq 0$ then $v \cdot xv > 0$
- ▶ positive semi-definite: If $\forall v \neq 0$ then $v \cdot xv \geq 0$
- ▶ The matrix A is **orthogonal** if $A^T = A^{-1}$. This is true iff all column vectors of A are orthonormal, and (equivalently) the row vectors are too. All eigenvalues of an orthogonal matrix are 1.
- ▶ If X is square and non-degenerate (distinct eigenvalues), its eigenvectors U form an **orthonormal basis**.

Projections are idempotent

- ▶ Once you project a vector into a subspace, projecting it again does nothing. Such projections P are called **idempotent**:

$$P^K = P, \quad \forall K \geq 1$$

- ▶ Spectral projections have this property.

Discussion

- ▶ How do we determine the elbow in a scree plot? Can we find examples with a discussion?
- ▶ Are any of the criterion “right”? When?
- ▶ Can we use cross-validation to obtain something more concrete? What would that look like?
 - ▶ Do we need a classification or similar supervised task? Is PCA useful for that?

R commands for matrix operations

```
A %*% B # matrix multiplication  
t(A) # matrix transpose  
diag(A) # diagonal vector of A  
diag(x) # a diagonal matrix formed of the vector x  
det(A) # determinant  
sum(diag(A)) # trace  
solve(A) # matrix inverse  
eigen(A) # Eigenvalue decomposition  
svd(A) # Singular Value Decomposition
```